

Lesson 4-3 · Period 1

Rational Expressions — Equivalent & Simplify

Klmsara-adapted · Student-centered · No Blooket

DOK 1

5 min

notice & wonder

Graph of $y = x + 2$ Consider the graph of the function $y = x + 2$.

1. What is the domain of this function?
2. Sketch a version that **excludes** $x = 2$ from the domain.
3. What kind of function might have a graph like that? Explain.

Write 1 **notice** + 1 **wonder** on your packet. One sentence: what is the equation doing at that skipped x -value?

DOK 1–2

12 min

teacher models

Equivalent + Simplify Rules (keep on your page)

1. **Factor** the numerator AND denominator completely.
2. Cancel only common **FACTORS** — never terms across $+$ or $-$.
3. State domain restrictions: every zero of the **ORIGINAL** denominator is excluded — including factors you canceled.

Common Error: Do NOT cancel terms that are added or subtracted across a fraction bar. Factor completely **FIRST**, then cancel only identical factor pairs.

Key insight: Canceling a factor never erases the domain restriction it imposed.

Explore — Think-Pair-Share (33 min)

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DOK 1–2

33 min

student work

7 items in order. Plan ~4–5 min per item. Cut #19 if pace slows.

Try It 1 · Equivalent expression Write an expression equivalent to $(x - 4)/x$ over domain $\{x \mid x \neq 0 \text{ or } -2\}$.

Practice #20 · Equivalent over restricted domain Write an equivalent expression: $x(x - 2)/(x - 5)$ for all x except 5 and -6 .

Practice #22 · Simplify + domain Simplify $(y^2 - 5y - 24)/(y^2 + 3y)$. State the domain.

DOK 1–2

33 min

student work

Try It 2 · Simplify by factoring and canceling Simplify each expression and state the domain.

a. $(x^2 + 2x + 1)/(x^3 - 2x^2 - 3x)$

b. $(x^3 + 4x^2 - x - 4)/(x^2 + 3x - 4)$

Practice #17 · Construct Arguments Explain how you can tell whether a rational expression is in simplest form.

Practice #19 · Domain restriction (extension) If the denominator is $x^3 +$

DOK 2

5 min

academic conversation

Sentence frame · Domain & Simplify “The factored form is _____, so the domain excludes $x =$ _____. After canceling the factor _____, the simplified expression is _____ with the **same** domain restriction.”

Bridge to Period 2 Next class: multiply and divide rational expressions. The same factor-and-cancel logic applies — plus you'll work with products and quotients.

Key rule carries over: always state domain restrictions before simplifying.

Exit Ticket — Summary Recap

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DOK 1–2

2 min

summary recap — not graded

Complete on your exit ticket To simplify $(x^2 + 3x - 10)/(x^2 - 4)$ I first factor to _____.

The restricted domain values are $x \neq$ _____.

After canceling the factor _____, my simplified expression is _____.

One biggest thing I learned today: _____.